

AI Engineer - Agent Screening Test

Practical assessment · Estimated effort: 1-2 days

Objective

Build a working AI agent that reads the provided vendor SKU and sales file and automatically produces e-commerce ready outputs: product content, lifestyle images, competitor benchmarking and a sales analysis.

What we are assessing is the **automation**. We want a pipeline that runs end to end from a single input file. Outputs assembled by hand, or by pasting rows into a chat window one at a time, will not score.

Input

One file is provided with this test: `test_data.xlsx`. Your agent must read this file directly. Do not retype, clean or restructure it by hand before running the pipeline — any cleaning must happen inside the pipeline.

The workbook has two sheets.

Sheet 1 — vendor data (25 SKUs, furniture)

Field	Notes
SKU, Masked SKU	Product identifier
Vendor1 Name, Brand	Supplier and brand
Category, Color, Material Used, Assembly Required	Core product attributes
Bullet 1 - Bullet 20	Raw vendor copy — unstructured, inconsistent, many blanks
Image URL 1	Product image used for the lifestyle image step
Weight / Length / Width / Height / Dim (1-8)	Product and shipping dimensions per carton

Sheet 2 — sales data (~10,700 order lines, Jul-Sep 2025)

Field	Notes
orderDate, orderId, invoiceNumber, customerReferenceNumber	Order identifiers
itemName, Masked SKU	Item sold
quantity, price, amount	Line quantity and value
Customer Name, orderStatus, shippingCarrier	Order context
Vendor, vendorType	Supplier and fulfilment route

Read the data before you build. The two sheets do not share SKUs — the 25 vendor SKUs do not appear in the sales file. Content, image and competitor work therefore runs on the vendor sheet; the sales analysis runs on the sales sheet as its own dataset. Order statuses include cancelled and pending lines, carrier values include unmapped entries, and the bullet columns are ragged. Handling this cleanly, and telling us how you handled it, is part of the test.

Agent Requirements

The agent takes the input file and performs the following four tasks automatically.

1. Content Generation

For each SKU in the vendor sheet, generate:

- Product title
- Product description
- 5 key features
- Important attributes (colour, material, assembly, dimensions and anything else worth surfacing on a listing)

Copy must be derived from the vendor bullets and attribute columns, not invented. Keep it marketplace ready: no placeholder text, no truncated sentences, no hallucinated specifications.

2. Lifestyle Image Generation

Using the product image URL supplied for each SKU:

- Generate a lifestyle image, or automate the image generation process end to end
- Keep the actual product appearance reasonably consistent with the source image

Images must be produced by the pipeline and saved to an output folder, named by SKU. If generation cost is a constraint, run a clearly stated subset and make the code capable of the full set.

3. Competitor Analysis

For a selected set of SKUs, identify comparable products and report:

- Competitor product
- Marketplace or website
- Competitor price
- Our price
- Price difference
- Short competitive insight

Our price is not given in the vendor sheet. Derive a reference price from the sales data, or state clearly how you sourced or assumed it.

4. Sales Analysis

Using the sales sheet, produce:

- Total sales
- Units sold
- Top performing SKUs
- Low and non-moving SKUs
- 3 to 5 business insights

State how you treated cancelled and pending orders, and anything else you filtered out.

Expected Agent Flow

Upload vendor file → analyse product → generate content → generate lifestyle image → research competitors → analyse sales → generate final output

The whole chain must run from one command, one workflow trigger or one upload. A human should not have to move data between steps.

How You Build It

Either approach below is accepted. Pick one and say which in your README.

Track A — Code-based agent (preferred)

Python or TypeScript, using an agentic framework such as LangGraph, LangChain, CrewAI, AutoGen, or a comparable alternative. Model and image-generation provider are your choice.

We expect: clear separation of steps or nodes, structured outputs, retry and error handling for API failures, and a single entry point (for example `python main.py --input test_data.xlsx`).

Track B — n8n workflow

A complete n8n workflow that performs the same four tasks. Submit the exported workflow JSON so it can be imported and run.

We expect: the file read and all four tasks inside the workflow, looping over SKUs rather than a fixed handful of hard-coded nodes, working error branches, and any Code nodes included in the export. Credentials must be referenced, never hard-coded.

Whichever track you choose, the same bar applies: one trigger runs the whole thing, and the pipeline works on new rows without being rewired.

Submission

Submit everything through **Google Drive**. Create one folder in your own Google account, named `FirstName_LastName_AI_Agent_Test` , set it to *Anyone with the link - Viewer*, and send us that single link.

The folder should contain:

Item	Notes
Working agent	Runnable, with setup steps and any environment variables listed (code track), or an importable workflow JSON (n8n track)
Source code	Place it in the folder, or include a GitHub link in the README
<code>/output</code>	Final output file — content, competitor and sales results
<code>/images</code>	Generated lifestyle images, named by SKU
README	Approach, track and framework used, how to run it, and the assumptions you made about the data

Do not email files as attachments and do not send a zip. One Drive link, checked from an incognito window before you send it, so we know the sharing actually works.

What We Look At

- Agent workflow and automation
- Content generation quality
- Competitor analysis
- Lifestyle image automation
- Sales analysis
- Code quality and explanation

The main thing we assess is whether you have built a real automated agent, rather than producing the final outputs by hand.